Caron 2020 ALL — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates all 8 samples; sample-iLISI rises 1.0→4.1 of a 6.85 ceiling (plateau θ 5–10). The manifold resolves into a cycling lobe and a low-cycling embryonic lobe; the cycling lobe is SCOPED OUT (25,788→19,800). On the scoped embryonic manifold samples and subtypes intermix on one connected component — GO.

Gate 2 — module geography → GO (non-branching)

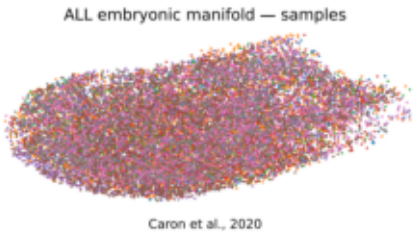
On the scoped manifold gastrulation is high at the root and falls along pseudotime (ρ=−0.65); Crest rises to a sharp inverse terminal gradient (decile t=0.97) opposing Gast (r=−0.74). Gast/pSom cross-repress (r=−0.55); pSom sits mid-axis (decile t=0.45). No resolution into two module-distinct arms — non-branching GO.

Gate 3 — cross-repression → NA

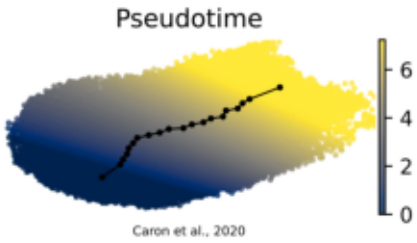
Non-branching topology: on the scoped embryonic manifold the PPT tree yields 2 tips / 0 forks (Gast-high root, Crest-high terminus) — no Gast/pSom co-expression hub. A clean fold-free curve (self-approach 3.0) is fit — no bifurcation to test — NA.

STEP FIGURES

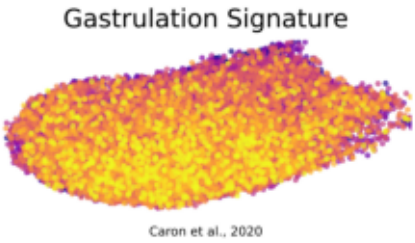
Manifold — samples



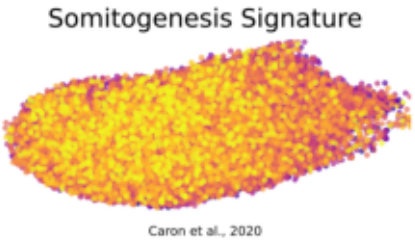
Pseudotime



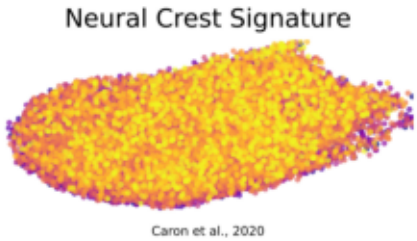
Gastrulation



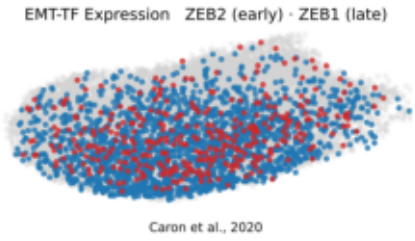
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Pediatric B/T-cell acute lymphoblastic leukemia (Caron et al., 2020), 10x — 19,800 embryonic blasts (cycling lobe scoped out) across 8 samples / 3 subtypes (ETV6-RUNX1, HHD, T-ALL). Non-branching Linear.

Verdict.
Pattern reproduces (non-branching / Linear) after scoping out the cycling lobe. A single continuous gastrulation → neural-crest embryonic axis on the 19,800-cell embryonic manifold, rooted at the Gast-high pole, with organised Gast/pSom cross-repression (r=−0.55) and a sharp terminal Crest gradient (decile t=0.97). The proliferation confound is resolved (pt↔proliferation ρ +0.62→+0.21). No bifurcation — Gate 3 NA.

Parameter choices

Min-cell threshold	≥100 (smooth counts) → 8/8 samples, 25,788 malignant cells	**Integration**	Harmony θ=10; n_pcs=20, n_neighbors=30, spread=1.3; 1,332 HVG
Compartment scoping	cycling lobe (86.5% cycling) scoped out → 19,800 embryonic cells	**Geometry**	Linear (non-branching); scFates curve N=20, fold-free (3.0); 2 tips / 0 forks
Root	curve tip at Gast-high, low-cycling origin (Gast/pSom early, Crest late)	**Modules recovered**	Gast 66/81 · pSom 77/98 · Crest 93/104
EMT-TFs	ZEB2 (early/root) → ZEB1 (late/terminus); SNAI1 also expressed	**Confound resolved**	pt↔proliferation ρ +0.62→+0.21; continuity |corr(t,DC1)|=0.91